LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Anti-skid devices, purpose and operating principle of anti-skid unit.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Anti-skid devices, purpose and operating principle of anti-skid unit.
5. Specific Objectives	17.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of anti-skid system diagrams in IB; Instructor supervision required for diagram tracing.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear?
(b) Why is braking necessary during the landing roll?
(c) What happens if wheel rotation exceeds the limit during landing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of anti-skid devices • Construction of typical landing gear units • Role of wheel speed sensors in anti-skid • Hydraulic pressure modulation mechanism • Prevention of wheel lock-up and skidding	L & Ds	PPT, IB	15 min
(b)	• Operating principle of anti-skid unit • Signal processing from wheel speed sensors • Comparison of wheel speed to reference speed • Activation of solenoid valves for pressure reduction • Restoration of braking pressure upon speed recovery	L & Ds	PPT, PDF	20 min
(c)	• Benefits of anti-skid during landing and takeoff • System limitations and failure modes • Integration with automatic braking systems • Maintenance checks for anti-skid components • Troubleshooting common anti-skid faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the anti-skid unit prevents wheel lock-up during landing.
(b) Describe the sequence of events when a wheel speed exceeds the threshold.
(c) How does the anti-skid system interact with the hydraulic braking system?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between manual braking and anti-skid assisted braking, referencing the next lesson plan on under carriage components.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Tracing and identification of various under carriage components, Shock observers, types, brief description and operation	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min